

Into the sky

A catapult officer, or “shooter,” aboard the USS *Theodore Roosevelt* gives a clear signal to the pilot of a F/A-18F Super Hornet after checking the wind speed and direction. Just below the deck lies the steam catapult needed to help launch the plane into the air.

The F/A-18F Super Hornet is more than **60 ft (18 m)** long and is used by the US military.



SKYWARD BOOST

AIRCRAFT CATAPULTS

An airplane normally requires a long runway to gain the speed required for takeoff. This is not possible when taking off from ships at sea, so the decks of aircraft carriers are equipped with powerful catapult systems that drag the planes along, helping them go from 0 to 150 mph (a standing start to 250 km/h) in just two seconds. Most aircraft carrier catapults are powered by the pressure exerted by superheated steam.

STEAM PRESSURE

The amount by which a force is concentrated or spread out is called pressure. The pressure to power an aircraft catapult is created by heating water. This turns it into steam (a gas) and causes its molecules to move very fast. These fast-moving molecules exert a force on the container they are in, pushing on the piston.

